

MUHAMMAD TAHA NASIR

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SUMMARY

Computer Science student at FAST-NUCES specializing in Generative AI, LLM applications, and intelligent software systems. Experienced in building AI-powered applications including Retrieval-Augmented Generation (RAG) systems, voice agents, and machine learning solutions using LangGraph, LangChain, FastAPI, React, and Python. Strong foundation in machine learning, backend engineering, and scalable software development.

EDUCATION

Bachelor of Science in Computer Science, FAST-NUCES

2023 – Present

Relevant Coursework: Artificial Intelligence, Machine Learning, Data Structures and Algorithms, Database Systems, Software Engineering, Linear Algebra, Probability

TECHNICAL SKILLS

Programming:	Python, C++, JavaScript, SQL, Bash
Machine Learning:	PyTorch, TensorFlow, Scikit-Learn, OpenCV, Pandas, NumPy, MLflow
Generative AI:	LLMs, GPT APIs, Llama, RAG, LangChain, LangGraph, LlamaIndex AI Agents, Prompt Engineering, Embeddings, Vector Databases
Backend:	FastAPI, Django, Flask, REST APIs, PostgreSQL, MongoDB, Neo4j
AI Tools:	Hugging Face Transformers, Model Evaluation, Semantic Search
Cloud & DevOps:	Docker, AWS, GitHub Actions, CI/CD, Deployment Basics
Frontend:	React, Next.js, Tailwind CSS

EXPERIENCE

Voice AI Engineering Intern, Verxeon

Jul 2026 – Present

Islamabad, PK (Hybrid)

- Developed AI voice agents using LLMs, LangGraph, and Retrieval-Augmented Generation (RAG) pipelines for automated conversational workflows.
- Integrated Twilio APIs, Google Calendar, and backend services to support real-time voice interactions and scheduling automation.
- Built modular AI workflows connecting speech recognition, LLM reasoning, tool execution, and response generation.

Enterprise Security Intern, Central Depository Company of Pakistan (CDC)

Jul 2026 – Present

Karachi, PK

- Supported enterprise security operations involving Identity and Access Management (IAM), governance, risk management, and compliance.
- Gained practical exposure to security controls, audit processes, and enterprise technology infrastructure.

Software Engineering Intern, NUSyS Lab

Jul 2026 – Sep 2026

FAST-NUCES, Hybrid

- Developed the Student Society Management Module for NUtomate, a university digital management platform.
- Implemented features for society management, elections, events, financial workflows, and administrative operations.
- Collaborated with faculty stakeholders to convert requirements into maintainable software solutions.

Peshawar, PK

- Assisted students in Expository Writing by providing feedback on technical communication, structure, and analytical reasoning.
- Evaluated assignments and guided students in improving documentation and problem-solving approaches.

Frontend Engineering Intern, hwhelp.me

Aug 2025 – Sep 2025

Doha, Qatar (Remote)

- Built responsive frontend components using HTML, CSS, Bootstrap, and JavaScript.
- Integrated PHP and MySQL backend functionality for dynamic content delivery and user management.

AI/ML Engineering Intern, Advtrix

Jun 2025 – Aug 2025

Karachi, PK

- Developed Python-based data processing pipelines using Pandas and Scikit-Learn for machine learning workflows.
- Applied K-Means clustering for customer segmentation, reducing manual analysis effort by 40%.
- Used MLflow for experiment tracking and improving reproducibility of machine learning experiments.

AI Software Engineering Fellow, Headstarter AI

Jun 2024 – Jul 2024

Remote

- Selected for a competitive AI engineering fellowship focused on building and deploying AI applications.
- Developed Generative AI applications using OpenAI APIs, FastAPI, Pinecone, React, and Docker.
- Implemented RAG pipelines, semantic search systems, and automated deployment workflows.

PROJECTS

AI Dental Receptionist: Voice AI Agent

Built an AI-powered voice assistant using Python, Pipecat, and LangGraph. Designed a real-time speech-to-speech workflow integrating STT, LLM reasoning, and TTS with WebRTC streaming, Google Calendar integration, and Chroma-based RAG search for automated appointment scheduling.

RescueAI: AI Emergency Response Platform

[GitHub](#)

Built a real-time emergency assistance platform using FastAPI and React. Implemented Llama-powered conversational AI, GPS-based SOS workflows, voice interaction, and asynchronous backend services for low-latency communication.

Veriflow: Revenue Leakage & Invoice Auditing System

[GitHub](#)

Developed a machine learning compliance system analyzing 5,500+ transactions worth over \$21M. Applied anomaly detection techniques to identify 86 freight-overcharge cases and automate revenue leakage analysis.

ApexKV: High-Performance In-Memory Cache Engine

[GitHub](#)

Implemented a Redis-inspired cache engine in Java featuring TTL expiration, Write-Ahead Logging (WAL), eviction policies, Pub/Sub messaging, TCP networking, and a monitoring dashboard.

Genetron: 5G Tower Placement Optimizer

[GitHub](#)

Developed an AI optimization platform using NSGA-II genetic algorithms and signal propagation modeling for 5G base station placement. Built FastAPI services and React visualizations achieving 99.4% simulated coverage.

CERTIFICATIONS

- **Generative AI Application Developer**, NCEAC-HEC & ULEF
- **Deep Learning Specialization**, DeepLearning.AI

- **Machine Learning Specialization**, Coursera / Stanford
- **NAVTTC Artificial Intelligence Program**, Government of Pakistan

ACHIEVEMENTS

- **ICPC Regional Qualifier** – Ranked 86th in Asia in the International Collegiate Programming Contest Qualification Round.
- **Top Performer** – Generative AI Application Developer Program (Final Score: 96.04%).
- Winner – CodeJail Competitive Programming Championship, FAST-NUCES Islamabad (2025).
- 3rd Position – FAST Problem Solving Competition (2023).

LEADERSHIP & ACTIVITIES

President, FAST Innovation Club 2025 – Present

- Lead club operations, coordinate technical initiatives, and collaborate with students, faculty, and industry partners.
- Organize technical workshops, innovation events, and programming competitions.
- Mentor students in AI, software development, and competitive programming.